

Differential Privacy Definition

Please watch [this short video](#) created by Google to get an overview of DP.

Differential privacy is a property of a data-processing algorithm, which ensures that the algorithm's output is roughly the same with or without the data of any single individual. The definition considers *neighboring datasets* that differ in the data of exactly one person. The definition says that $F(x)$ satisfies ϵ -differential privacy if: for all possible neighboring datasets x and x' and all possible outputs S of the function:

$$\frac{\Pr[F(x) \in S]}{\Pr[F(x') \in S]} \leq e^\epsilon$$

When ϵ is reasonably small, this definition means that the probability of observing the output S should be similar, whether you run F on x or x' . The smaller ϵ is, the closer these probabilities need to be.

Meaning of ϵ ; Composition

The parameter ϵ controls the strength of the privacy guarantee. A smaller ϵ means more privacy, and a larger ϵ means less privacy. The ϵ parameter is often called the *privacy parameter*, the *privacy loss*, or the *privacy budget*. Typical ϵ values are between 0 and 10, with a preference for low single-digit numbers.

If two differentially private releases are made, one with parameter ϵ_1 and the other with parameter ϵ_2 , the total privacy loss is $\epsilon = \epsilon_1 + \epsilon_2$. This property is called *sequential composition*, and it allows multiple analyses to be performed on the same data while bounding the total privacy harm.

Differential Privacy Mechanisms and Sensitivity

One way to satisfy differential privacy is by adding carefully designed random noise to the result of a data analysis. More noise yields better privacy (lower ϵ), while less noise yields worse privacy (higher ϵ). A common method is to add noise drawn from the Laplace distribution, with the scale of the noise dependent on the desired privacy parameter ϵ and the *sensitivity* of the analysis. Sensitivity is a number that measures how much the analysis result can change between neighboring datasets; for analyses with larger sensitivities, more noise is required to achieve a desired differential privacy guarantee.

Mechanisms like the Laplace mechanism can be used to build differentially private versions of common data analytics tasks. For example, a differentially private count, sum, or average can

be obtained by first performing the standard aggregation function and then adding Laplace noise. Libraries for differential privacy typically include built-in mechanisms for specific tasks designed this way; some libraries determine the sensitivities for these tasks automatically, while others require the programmer to provide the sensitivity values.